

Chayce Reed

HSE 751 – Programming for Health Data Science

Week 2 - Reproducibility Exercise: Diabetes Risk Factor Analysis

09/30/26

Several reproducibility issues were identified with the original notebook. First, the package versions were not specified. This means that results could change depending on the time and the environment that the notebook is run. For example, the `scipy` package was updated and the `ttest_ind()` function may behave differently or expect different arguments than when the notebook was first developed/validated. Second, the random sample and bootstrap operations in the reproducibility check code blocks did not have a seed defined, so they gave different results each run. Third, the dataset had to be uploaded manually before running. Fourth, file paths and labels were typed inline wherever they were used, instead of defined once upfront for reusability. Fifth, the cleaned dataset and figures were not saved after running the notebook. Sixth, there were limited inline comments and markdown instructions throughout the notebook.

The first issue was addressed by specifying and installing the version of the packages that were used during the creation and validation of the notebook. This ensures that all future runs will use the same package versions, regardless of what the default Colab environment uses. A package version check was also included to confirm the correct versions were installed and are being used. The second issue was addressed by defining a random seed in the setup steps and passing it to every random operation throughout the notebook (i.e., random sample and bootstrap operations in the reproducibility check). The third issue was addressed by having the notebook check if the dataset was already uploaded to the environment, and if not, automatically pull it from my public GitHub repository and save it in the local environment for future runs. The fourth issue was addressed by defining the file paths, column groups, and display labels once upfront, and reusing those throughout the rest of the notebook. This allows easy access and updating for future runs. The fifth issue was addressed by saving the cleaned dataset and each figure to the defined output location in the environment. This allows for the dataset and figures to be exported, shared, or compared for future runs. Finally, the sixth issue was addressed by adding inline comments to explain code or specific decisions, a markdown description of each major section and the specific subsections, and a README containing the Google Colab link, data source and provenance, package requirements, and instructions for running the notebook. These reproducibility changes were verified by restarting the Colab runtime and rerunning the full notebook, both with the dataset uploaded and without it, and verifying the reproducibility check outputs after each run.